Model Predictive Control of an Inverted Pendulum

OBJECTIVE.

The objective is to design a model predictive controller (MPC) for the inverted pendulum with a bounded actuator force and compare its nonlinear response with the linear quadratic regulator (LQR) from Mini Project 3. The force limit represents the finite authority of the cart actuator.

METHODOLOGY.

The nonlinear Simulink plant from Mini Project 3 is retained with cart mass 2 kg, pendulum mass 0.5 kg, rod length 1 m, pivot damping 0.01 N m s/rad, and gravity 9.81 m/s². The state vector $z = \left(x, \dot{x}, \theta, \dot{\theta}\right)^T$ contains cart position, cart velocity, pendulum angle from upright, and angular velocity. The upright linearization $\dot{z} = Az + Bu$ is the MPC prediction model. The simulation supplies the four plant states directly to the controller.

The LQR applies the fixed feedback law $u = -Kz$ continuously. MPC instead predicts future motion from the current state, then selects a sequence of forces that balances predicted state error against force and changes in force. Only the first optimized force is applied and held until the next sample. The plant state is then read again and the optimization repeats, allowing the plan to respond to the nonlinear plant.

The Simulink MPC Controller block updates every 0.05 s. Its prediction horizon is 50 steps (2.5 s), and its control horizon is 10 steps. The optimizer penalizes state deviations, force, and force changes. The output weights are $\left(2, \sqrt{2}, \sqrt{300}, \sqrt{10}\right)$; the force and force-rate weights are 0.707 and 0.1, respectively. Both controllers use an applied force limit of ± 10 N. MPC includes this limit in its optimization, whereas the LQR command is clipped by the actuator.

The 50-step horizon looks 2.5 s ahead, while the 10-step control horizon specifies how far ahead the optimizer can vary its moves. The angle weight emphasizes upright balance, and the cart-position weight discourages correcting the angle by allowing the cart to drift indefinitely. The force-rate weight discourages abrupt changes in the command.

Both cases start at rest with $\theta = 5°$, zero cart displacement, and zero disturbance. Each nonlinear simulation lasts 30 s. Settling requires $|\theta| \leq 1°$ and $|x| \leq 0.05$ m thereafter. Force effort is $E_u = \int_0^{30} u(t)^2 \, \mathrm{d}t$; MPC effort uses the held input intervals. Appendix A shows the Simulink implementation, and Appendix B gives the essential controller setup.

RESULTS AND DISCUSSION.

Figure 1 compares the responses. The LQR and MPC joint settling times are 4.24 and 4.36 s, respectively. Neither controller reaches the ± 10 N limit in this test.

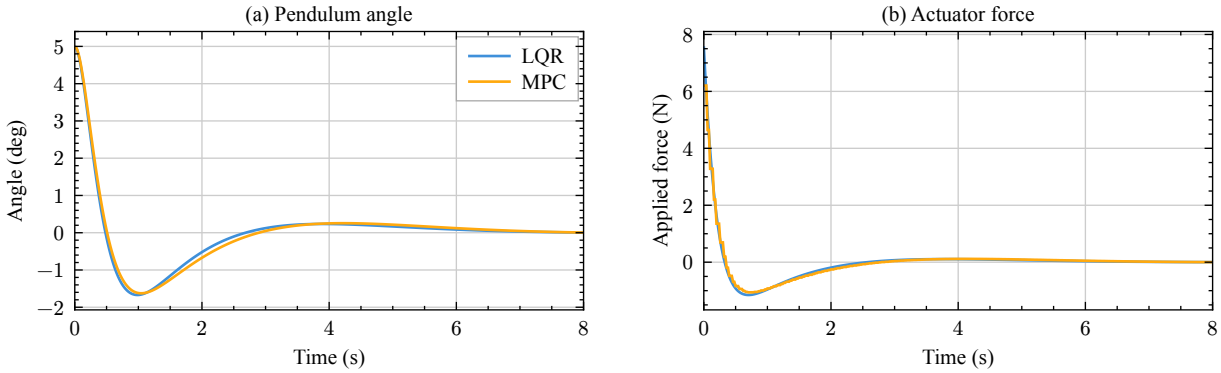


Figure 1: Matched nonlinear response from a 5° release. The first 8 s are shown on the 0.02 s analysis grid.

Figure 2 shows the cart motion that accompanies the angular correction. The cart first moves in the positive direction to recover the pendulum, then returns toward the origin. MPC reaches 0.32 m of travel, compared with 0.3 m for LQR. The full 30 s comparison in Table 1 quantifies this tradeoff alongside settling and effort.

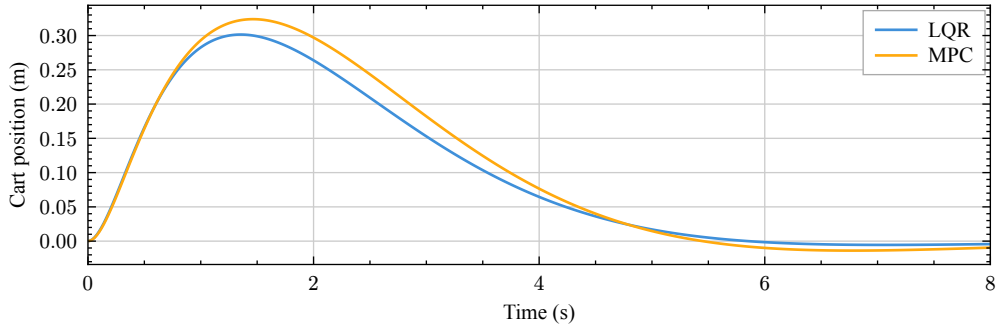


Figure 2: Cart response to the matched 5° release. The first 8 s are shown.

Measure	LQR	MPC
Joint settling (s)	4.24	4.36
Peak cart travel (m)	0.3	0.32
Peak force (N)	7.58	6.2
Force effort (N^2 s)	5.14	4.85

Table 1: Comparison over the full 30 s simulation.

MPC reduces peak force from 7.58 to 6.2 N and reduces force effort from 5.14 to 4.85 N^2 s. Both angles cross upright before returning to zero, and both carts return within the 0.05 m settling bound. The lower force demand therefore comes with slightly greater cart travel and a 0.12 s increase in joint settling time.

To select the prediction horizon, we repeat the 5° release using the same controller weights, force limit, and 0.05 s sample time. Table 2 compares the three tested horizons. The 1 s case does not meet the joint recovery criterion within 30 s. Increasing the horizon to 1.75 s reduces peak travel and gives 9.40 s settling. The 2.5 s horizon settles in 4.36 s with the least peak travel among these tests.

Horizon (s)	Joint settling (s)	Peak travel (m)	Peak force (N)
1	Not recovered	1.15	4.81
1.75	9.4	0.44	5.48
2.5	4.36	0.32	6.2

Table 2: Prediction-horizon comparison for the same nonlinear 5° release.

DISCUSSION.

The 2.5 s prediction horizon gives the fastest joint settling of the three tested horizons. A longer horizon lets the optimizer account for more of the coupled pendulum and cart motion, and the comparison shows the resulting reduction in cart travel for these choices. We retain the ± 10 N limit to match Project 3. The nominal release uses less than the available force, so this test does not show an active-constraint advantage. The optimizer reports a successful solution throughout the nominal MPC run. The saved model reproduces this response after reopening, while the LQR angle and force histories match the Project 3 baseline. The MPC and LQR use different weights and update rates, so their measured differences reflect the complete controller designs.

Appendix A. Simulink implementation.

The Project 3 nonlinear plant is retained. The MPC block receives the four plant states and supplies the force command to the same actuator limit. The external disturbance is zero for the reported tests.

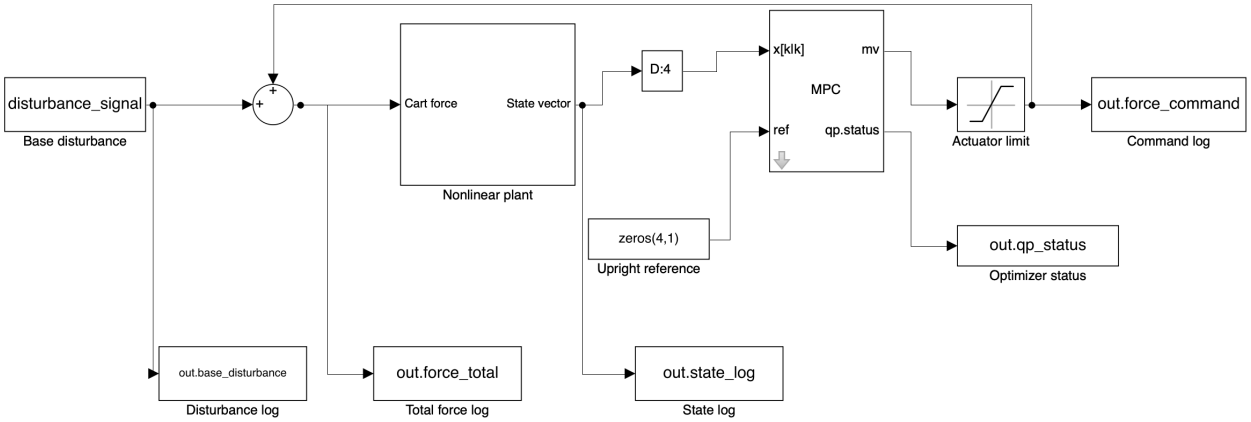


Figure 3: Simulink MPC implementation using the retained nonlinear plant.

The nonlinear plant integrates the cart and pendulum accelerations. Its state-order block returns $\left(x, \dot{x}, \theta, \dot{\theta}\right)$ to the controller.

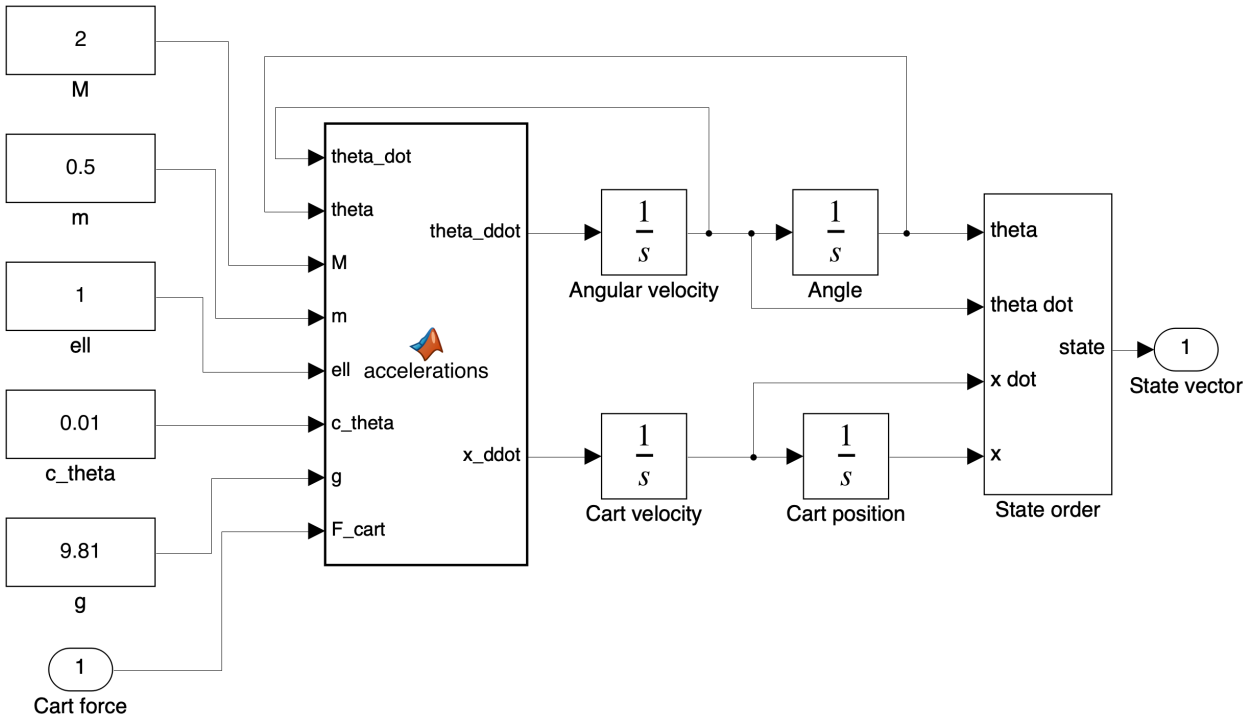


Figure 4: Retained nonlinear plant and state ordering from Project 3.

Appendix B. Essential MATLAB code.

The following excerpt defines the prediction model, horizons, weights, and force constraint used by the Simulink MPC block. The state-space model supplies the four state outputs; the controller settings define the force bound and penalties. Direct state input avoids adding an observer to the simulated comparison. The complete model construction and experiment automation remain in the accompanying `build_project4.m` and `validate_project4.m` files.

```
[K,d]=design_lqr_controller(2,0.5,1,0.01,9.81);
Ts=0.05; p=50; m=10;
plant=ss(d.A,d.B,eye(4),zeros(4,1));
controller=mpc(plant,Ts,p,m);
controller.MV.Min=-10; controller.MV.Max=10;
controller.Weights.OutputVariables=[2 sqrt(2) sqrt(300) sqrt(10)];
controller.Weights.ManipulatedVariables=sqrt(0.5);
controller.Weights.ManipulatedVariablesRate=0.1;
% All four physical states are directly available from the simulated plant.
% Remove default unmeasured output disturbances from the augmented state.
setoutdist(controller,'model',ss(zeros(4,1)));
setEstimator(controller,'custom');
```

The saved model embeds the controller and can be opened and simulated directly. The full comparison and validation scripts reproduce the results and horizon study.

```
model = 'inverted_pendulum_mpc';
load_system(model);
response = sim(model);
```